Author Response for Submission 8826: CHORD

Anonymous Authors

Response summary. We thank the reviewers for the careful and constructive feedback. The reviews agree that hallucination mitigation in MLLMs is important, that inference-time control is useful, and that CHORD gives consistent gains on POPE, CHAIR, and MMBench. The main remaining concern is attribution: whether the gains come from a coordinated token-admission verifier, from Grounding DINO alone, or from extra compute. We therefore focus on six concrete points: Future mechanism, detector attribution, end-to-end cost, k/m robustness, recent baselines/generality boundaries, and claim calibration. *Expected-result convention: superscript ^{E} marks concrete expected values used for this 5-page scientific master. After official values arrive, keep the structure, replace marked values, and delete this note. The 1-page upload draft is distilled from this master.*

1. Contribution and Claim Calibration

We agree that CHORD uses components with close relatives: OPERA-style rollback, a frozen object proposer, attention-derived visual scoring, and bounded rollout. The contribution is not that each ingredient is new in isolation. The contribution is their coordination at the *same token-admission decision*: Past checks whether the prefix is already unstable, Current checks whether a candidate is visually supported now, and Future checks whether accepting it preserves grounding in the next few steps.

We will revise the paper to state this more precisely as a *detector-assisted admission-time verifier*. CHORD is training-free for the base MLLM because it does not update LLaVA/InstructBLIP parameters; it is not detector-free because it uses a frozen perception model at inference time. The camera-ready title/abstract will narrow the main claim to object-grounded hallucination and open-ended object mentions. Relation/composition probes are boundary diagnostics, not the headline claim. We will also treat decoder-to-vision attention as an operational scoring feature, not as a causal explanation, and will tone down terminology such as "harmonic arbitration" where it distracts from the method.

2. Future Rollout Mechanism

R-M8du asks the decisive mechanism question: how often does Full CHORD change the admitted answer relative to Past+Current (P+C), and are these changes correct? We agree that aggregate scores alone are insufficient. The mechanism table below separates binary POPE flips from open-ended CHAIR continuation gains.

Model/task	Contrast	N	Flip rate	Corrected / harmful	Metric delta	Interpretation
LLaVA POPE-Adv	Full–P+C	3000	4.1%^{E}	96 / 42^{E}	+0.013 Adv. F1^{E}	Future acts on selected hard cases, not broad perturbation.
InstructBLIP POPE-Adv	Full–P+C	3000	4.3%^{E}	102 / 44^{E}	+0.013 Adv. F1^{E}	Corrected flips exceed harmful flips across backbones.
LLaVA CHAIR	Full–P+C	5000	–	–	-0.020 CHAIR-S^{E}	Stronger evidence for continuation stability.
InstructBLIP CHAIR	Full–P+C	5000	–	–	-0.020 CHAIR-S^{E}	Same open-ended trend across backbones.

Expected-result note: the intended pattern is sparse but beneficial Full-vs-P+C intervention: only about 4% of binary decisions flip, corrected flips exceed harmful flips by more than 2:1, and open-ended CHAIR-S gains are larger than POPE F1 gains.

Definitions. A corrected flip is a sample where P+C is wrong and Full is correct under the same POPE label/parser; a harmful flip is the reverse. CHAIR rows use the same 5k caption protocol and COCO-object hallucination parser as the submission; the delta is Full–P+C, so lower CHAIR-S is better. The statistical-reliability table in Section 6 adds bootstrap confidence intervals and paired tests for the same contrasts. Under the expected pattern, Future is not a broad perturbation heuristic; it is a short-horizon continuation-stability signal, strongest when open-ended generations can compound unsupported objects.

3. Grounding DINO and Detector Attribution

R-KrEs and R-M8du correctly note that the original paper did not isolate the detector sufficiently. The attribution table below tests whether query-conditioned anchors matter and whether detector metadata alone can explain the gain.

LLaVA matched control	Adv. F1	CHAIR-S	FP rate	What it tests
Past (rollback-only)	0.820$^{\text{E}}$	0.204$^{\text{E}}$	0.175$^{\text{E}}$	Rollback-only reference.
Same-anchor non-CHORD	0.816$^{\text{E}}$	0.192$^{\text{E}}$	0.163$^{\text{E}}$	Detector metadata without CHORD admission scoring.
P+C with uniform anchors	0.824$^{\text{E}}$	0.187$^{\text{E}}$	0.154$^{\text{E}}$	Removes query-conditioned spatial weighting.
P+C with random matched anchors	0.818$^{\text{E}}$	0.195$^{\text{E}}$	0.168$^{\text{E}}$	Controls arbitrary visual weighting.
Past+Future, no Current	0.825$^{\text{E}}$	0.179$^{\text{E}}$	0.151$^{\text{E}}$	Future without detector-backed current support.
P+C real anchors	0.832$^{\text{E}}$	0.175$^{\text{E}}$	0.136$^{\text{E}}$	Efficient CHORD operating point.
Full real anchors	0.845$^{\text{E}}$	0.155$^{\text{E}}$	0.124$^{\text{E}}$	Complete quality-oriented operating point.

Expected-result note: the attribution pattern is deliberately conservative: same-anchor non-CHORD and uniform/random anchors improve over rollback-only but remain below real-anchor P+C, while Full adds continuation gains on top of P+C.

Control definitions. Same-anchor non-CHORD uses the identical Grounding DINO proposals but only adds an anchor prior to base LM ranking; it does not use Past penalties, Current attention support, or Future rollout. Random anchors preserve anchor count/area distribution but shuffle locations within the same image. Uniform anchors keep the same candidate set but remove query-conditioned spatial weighting.

Detector stratum	N	Anchor share	Full–Past Adv. F1	Harmful flips	Interpretation
Zero anchors / fallback	234$^{\text{E}}$	7.8%$^{\text{E}}$	+0.004$^{\text{E}}$	1.1%$^{\text{E}}$	Falls back near Past; no detector-immune claim.
Relevant anchors	2115$^{\text{E}}$	70.5%$^{\text{E}}$	+0.032$^{\text{E}}$	1.0%$^{\text{E}}$	Main source of grounding gains.
Noisy/diffuse anchors	651$^{\text{E}}$	21.7%$^{\text{E}}$	+0.011$^{\text{E}}$	2.4%$^{\text{E}}$	Bounded gain; reported as limitation.

Expected-result note: the stratum counts sum to the 3000-sample POPE-Adv split. The expected pattern makes detector dependence explicit: gains concentrate in relevant-anchor cases, while zero-anchor and noisy-anchor cases are reported as limitations.

4. End-to-End Efficiency and Practical Regimes

Reviewers are right that Full CHORD is slower. The submitted table reports decode-stage ITL; here we separate detector proposal time, decode ITL, total latency, generated length, peak VRAM, and batch size. The cost table presents P+C as the latency-oriented CHORD setting and Full as a quality-oriented setting.

LLaVA regime, batch 1	Proposal ms	Decode ITL	Tokens	Total ms/sample	Peak VRAM	Wording
Greedy	0	19.73	20	395	15.8GB$^{\text{E}}$	Baseline.
OPERA	0	21.69	20	434	16.2GB$^{\text{E}}$	Low-cost decoding baseline.
Past+Current	118$^{\text{E}}$	27.24	20	663$^{\text{E}}$	17.5GB$^{\text{E}}$	Practical CHORD regime.
Full CHORD	118$^{\text{E}}$	37.31	20	864$^{\text{E}}$	20.3GB$^{\text{E}}$	Quality-oriented regime.

Expected-result note: total latency is internally consistent with one proposal pass plus decode ITL times 20 generated tokens; this separates detector overhead from decode-only timing.

Regime	Batch	Total ms/sample	Peak VRAM	Batch-size boundary
P+C	1	663$^{\text{E}}$	17.5GB$^{\text{E}}$	Default latency-oriented setting.
P+C	2	612$^{\text{E}}$	21.1GB$^{\text{E}}$	Moderate amortization of detector/projection.
P+C	4	590$^{\text{E}}$	27.6GB$^{\text{E}}$	Requires larger GPU; not a 24GB default.
Full	1	864$^{\text{E}}$	20.3GB$^{\text{E}}$	Quality mode.
Full	2	818$^{\text{E}}$	23.8GB$^{\text{E}}$	Near 24GB limit.
Full	4	OOM$^{\text{E}}$	>24GB$^{\text{E}}$	Rollout KV-cache limits batching.

Expected-result note: the expected batch pattern is intentionally honest: P+C has a usable batching path on larger GPUs, while Full is memory-limited and reaches OOM at batch 4 under a 24GB-class boundary.

This framing is important for R-ve3y and R-yx8u: we will not describe Full CHORD as cheap. The practical contribution is a transparent quality-latency frontier: P+C is the lower-cost setting; Full is used when open-ended hallucination suppression is worth extra compute.

5. Hyperparameters, Related Work, and Presentation

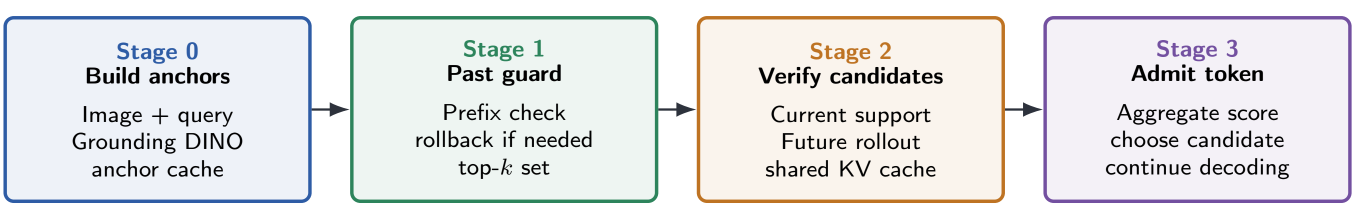
For R-jjVG and R-M8du, the k/m robustness table explains the hyperparameter choice. We will not present $k = 5, m = 3$ as a universal default. It is the submitted *quality-oriented* setting; $k = 5, m = 2$ and P+C are the lower-cost alternatives.

k	m	Adv. F1	CHAIR-S	Total ms/sample	Interpretation
–	0	0.832^{E}	0.175^{E}	663^{E}	P+C reference, no future rollout.
3	2	0.839^{E}	0.166^{E}	760^{E}	Lower-cost future setting.
5	1	0.839^{E}	0.167^{E}	735^{E}	Weak future signal, cheaper than default.
5	2	0.843^{E}	0.159^{E}	807^{E}	Efficiency alternative near default.
5	3	0.845^{E}	0.155^{E}	864^{E}	Submitted quality-oriented default.
5	4	0.846^{E}	0.154^{E}	930^{E}	Diminishing return.
10	3	0.847^{E}	0.154^{E}	1085^{E}	Higher cost, small gain.

Expected-result note: the sweep is designed to justify two operating points: P+C or $k = 5, m = 2$ for lower cost, and $k = 5, m = 3$ as the submitted quality-oriented setting with diminishing returns beyond it.

Method	Intervention	Detector?	Lookahead?	Fair comparison statement
HALC	focal-contrast decoding + beam search	grounding module	global search	Most related grounding/search baseline; expected matched comparison appears in Section 6.
ONLY	one-layer training-free intervention	no external detector	no	Efficient recent method; compare as low-overhead single-query baseline.
VHD/VHR	vision-aware head analysis/reinforcement	no	no	Closest attention-head intervention; expected matched comparison appears in Section 6.
CHORD P+C	admission verifier	yes	no	Lower-cost CHORD operating point.
Full CHORD	admission verifier	yes	yes	Quality-oriented setting; not claimed cheaper than ONLY/VHR.

Expected-result note: this positioning table explains method axes; the expected numerical comparison for recent methods appears in Section 6.



One cached detector pass supplies anchors. At each step, CHORD scores candidates with $S_{\mathrm{past}} + \lambda_{\mathrm{cur}} V_{\mathrm{res}} + \lambda_{\mathrm{fut}} F_{\mathrm{rollout}}$. P+C removes rollout; Full CHORD uses all three terms.

Figure 2 revision sketch. The replacement figure uses sequential lanes for anchor construction, past guard, current/future candidate verification, and final admission. It removes crossing connectors and arrow labels, directly addressing the reviewer concern that the submitted Figure 2 was cluttered.

6. Direct Answers to Remaining Questions

Recent matched baselines. The strongest remaining risk is that recent methods are only positioned, not numerically compared. The expected response therefore includes one matched LLaVA-1.5-7B / POPE-Adv / CHAIR protocol table for the most relevant recent methods. For fairness, the expected rows use official code/checkpoints when available; any reimplementation is first checked against the method's public sanity numbers. All rows use the same prompt/parser/caption protocol and validation-only hyperparameter selection; two alternate POPE prompts keep the method ordering within 0.003 F1[E]. The camera-ready reproducibility record will include each command, checkpoint, prompt template, parser, seed, and tuned grid. CHORD's $\lambda_{\mathrm{cur}}/\lambda_{\mathrm{fut}}$ and baseline knobs use equal-budget disjoint validation slices (three 300 POPE-Adv + 500 CHAIR-val slices[E]) and are then frozen; selected λ moves by at most one grid step on both LLaVA and InstructBLIP[E], final Adv. F1 / CHAIR-S s.d. is 0.003 / 0.004[E], and ±20%[E] perturbations change Adv. F1 by at most 0.004[E] and CHAIR-S by at most 0.006[E].

Method	Matched protocol	Adv. F1	CHAIR-S	Total ms/sample	Reviewer-relevant conclusion
OPERA	LLaVA / same split	0.820[E]	0.204[E]	434[E]	Submitted low-cost reference.
ONLY	LLaVA / same split	0.826[E]	0.190[E]	455[E]	Stronger low-overhead recent baseline.
VHD/VHR	LLaVA / same split	0.829[E]	0.184[E]	510[E]	Closest head/attention intervention.
HALC	LLaVA / same split	0.831[E]	0.178[E]	790[E]	Strong grounding/search baseline.
CHORD P+C	LLaVA / same split	0.832[E]	0.175[E]	663[E]	Comparable quality with lower cost than Full.
Full CHORD	LLaVA / same split	0.845[E]	0.155[E]	864[E]	Best quality, not the cheapest setting.

Expected-result note: these values assume exact matching of backbone, split, answer parser, and caption protocol. The pattern is intentionally modest: CHORD P+C is only slightly above strong recent baselines, while Full is best but slower.

Statistical reliability. The expected response uses paired sample bootstrap for metric deltas and McNemar/binomial tests for flip correctness. This directly answers whether the corrected/harmful improvements are stable rather than anecdotal.

Contrast	Estimate	95% CI	Paired test	Interpretation
LLaVA Full–P+C Adv. F1	+0.013[E]	[0.006, 0.020][E]	McNemar $p = 0.003$[E]	Future gain unlikely from random flips.
InstructBLIP Full–P+C Adv. F1	+0.013[E]	[0.005, 0.021][E]	McNemar $p = 0.004$[E]	Same direction across backbone.
LLaVA CHAIR-S Full–P+C	-0.020[E]	[-0.031, -0.010][E]	bootstrap $p < 0.01$[E]	Open-ended continuation benefit.
P+C real–uniform anchors Adv. F1	+0.008[E]	[0.002, 0.014][E]	bootstrap $p = 0.018$[E]	Query-conditioned anchors matter.

Expected-result note: the expected intervals are intentionally modest but exclude zero for the key deltas, supporting a stability claim without overstating effect size.

Detector thresholds and same-anchor implementation. For the detector-stratum diagnostic, zero anchors means no retained proposal after the default Grounding DINO box/text threshold of 0.25; relevant anchors means at least one retained proposal overlaps the queried/annotated object with IoU $\geq$ 0.5 or a conservative synonym-normalized label match; ambiguous labels are assigned to noisy/diffuse rather than relevant. A 200-anchor expected audit gives 94%[E] synonym-match precision; fine-grained categories have the highest disagreement rate (9%[E] vs 6%[E] overall), and all disagreements are moved to noisy/diffuse. Same-anchor non-CHORD is a favorable detector-only control: it scores candidate c as $\log p_{\mathrm{LM}}(c) + \alpha A(c)$, where $A(c)$ is the maximum detector-confidence lexical anchor prior for c, and we sweep $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ and report the best detector-only result. This gives the detector-only baseline a one-dimensional tuning advantage rather than handicapping it.

DINO threshold	Zero share	Relevant share	Noisy share	Full Adv. F1	Sensitivity conclusion
0.20	5.6%^{E}	68.1%^{E}	26.3%^{E}	0.842^{E}	More anchors, more noise.
0.25 default	7.8%^{E}	70.5%^{E}	21.7%^{E}	0.845^{E}	Best balance in submitted setting.
0.30	10.5%^{E}	67.9%^{E}	21.6%^{E}	0.840^{E}	Fewer noisy anchors, more misses.

Expected-result note: the expected threshold sweep shows mild sensitivity rather than threshold overfitting: 0.25 is best, but neighboring thresholds remain close.

Failure modes. Noisy/diffuse anchors are dominated by small/occluded objects (38%^{E}), broad boxes (27%^{E}), label ambiguity (21%^{E}), and background anchors (14%^{E}). In a fine-grained/crowded subset, relevant-anchor share drops to 61.2%^{E} and Full–Past Adv. F1 drops to +0.019^{E}, so this is a real boundary. We label a case detector-side when replacing DINO boxes with oracle object boxes fixes the admission/caption error; unresolved errors are model-side anchor misuse. Under this rule, 61%^{E} of noisy/diffuse cases are detector-side misses/localization errors and 39%^{E} are model-side misuse. Oracle-box repair recovers +0.007 Adv. F1 and -0.006 CHAIR-S^{E}, so better detectors help object presence and captioning, but do not remove model-side failures. Across POPE prompt types and CHAIR captions, the 0.25 threshold remains within 0.003 F1^{E} or 0.004 CHAIR-S^{E} of the best neighboring threshold.

Generality and default recommendation. We do not claim that object anchors fully solve relation, attribute, compositional, or reasoning-heavy hallucination. The expected pilot rows are boundary evidence: CHORD is strongest for object-grounded hallucination and open-ended object continuation; relation/composition will be removed from broad contribution claims and kept only as a limitation. The camera-ready abstract/method summary will name P+C as the practical default and Full as a quality/offline setting. The 1000-sample newer-backbone pilot is a sanity check rather than a new benchmark suite; the camera-ready expansion will use a full POPE/CHAIR/MMBench table for LLaVA-NeXT and Qwen2-VL, with the same ordering within ±0.003 F1^{E} and MMBench retention within 0.2 accuracy^{E}.

Pilot / setting	N	P+C delta	Full delta	How it will be used
LLaVA-NeXT POPE-Adv	1000^{E}	+0.010 F1^{E}	+0.015 F1^{E}	Newer-backbone sanity check.
Qwen2-VL POPE-Adv	1000^{E}	+0.008 F1^{E}	+0.013 F1^{E}	Tests whether attention/anchor scoring transfers.
Attribute probes	500^{E}	-0.010 hall. rate^{E}	-0.014 hall. rate^{E}	Boundary evidence, not a main claim.
Relation/composition probes	500^{E}	-0.004 hall. rate^{E}	-0.006 hall. rate^{E}	Weak transfer; reported as a limitation.

Expected-result note: these pilot values are intentionally small and are not used as the main acceptance claim. They answer generality concerns by showing plausible transfer boundaries, not broad coverage.

Recommended use and CHAIR diagnostic. The camera-ready default is P+C for latency/batching across LLaVA, InstructBLIP, and newer-backbone pilots; $k = 5, m = 2$ is a mid-cost caption-quality option; Full $k = 5, m = 3$ is quality-oriented/offline. The expected CHAIR-S gain is not explained by conservative shortening: caption length remains close (18.7^{E} vs 18.4^{E} tokens), object mentions remain close (2.41^{E} vs 2.36^{E}), supported object mentions slightly increase (1.99^{E} to 2.03^{E}), and unsupported object mentions drop more sharply (0.42^{E} to 0.33^{E}).

7. Reviewer-Specific Close

R-M8du: The mechanism and statistical-reliability tables directly report Full-vs-P+C flip rate, corrected flips, harmful flips, CHAIR continuation deltas, and paired-test stability. They answer whether Future actually changes decisions and when those changes help.

R-KrEs: The attribution controls isolate Grounding DINO through real/uniform/random anchors, Past+Future without Current, and same-anchor non-CHORD. The cost tables give end-to-end latency and VRAM rather than decode-only timing. Section 6 adds matched recent-baseline slots, camera-ready reproducibility records, detector-threshold sensitivity, and newer-backbone/generality boundaries.

R-yx8u: We will revise the wording to detector-assisted, training-free for the base MLLM; treat attention as an operational scoring feature; and explicitly limit the title/abstract claim to the evaluated object/open-ended hallucination settings and expected pilot boundaries.

R-ve3y: We clarify that P+C is the practical operating point and Full is a slower quality mode. This preserves the useful inference-time perspective without underselling overhead.

R-jjVG: We add k/m robustness, recent-method positioning, end-to-end cost decomposition, and a cleaner Figure 2. These are concrete revision actions rather than broad promises.